

## Problem

Find the inverse Fourier transform of:

a.  $H(\omega) = \frac{6(3 + j2\omega)}{(1 + j\omega)(4 + j\omega)(2 + j\omega)}$

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b.  $Y(\omega) = \pi\delta(\omega) + \frac{1}{j\omega} + \frac{2(1 + j\omega)}{(1 + j\omega)^2 + 16}$

## Step-by-step solution

## Step 1 of 7

(a) The given function is

$$H(\omega) = \frac{6(3 + j2\omega)}{(1 + j\omega)(4 + j\omega)(2 + j\omega)}$$

Using partial fraction expansion,

$$\frac{6(3 + j2\omega)}{(1 + j\omega)(4 + j\omega)(2 + j\omega)} = \frac{A}{1 + j\omega} + \frac{B}{4 + j\omega} + \frac{C}{2 + j\omega}$$

$$6(3 + j2\omega) = A(4 + j\omega)(2 + j\omega) + B(1 + j\omega)(2 + j\omega) + C(1 + j\omega)(4 + j\omega)$$

## Step 2 of 7

Put  $j\omega = -1$ ,

$$6(3 - 2) = A(4 - 1)(2 - 1) + B(1 - 1)(2 - 1) + C(1 - 1)(4 - 1)$$

$$6(1) = A(3)(1) + 0 + 0$$

$$3A = 6$$

$$\therefore A = 2$$

## Step 3 of 7

Consider  $j\omega = -2$

$$6(3 - 4) = A(4 - 2)(2 - 2) + B(1 - 2)(2 - 2) + C(1 - 2)(4 - 2)$$

$$6(-1) = 0 + 0 + C(-1)(2)$$

$$-6 = -2C$$

$$\therefore C = 3$$

## Step 4 of 7

Consider  $j\omega = -4$

$$6(3-8) = A(4-4)(2-4) + B(1-4)(2-4) + C(1-4)(4-4)$$

$$6(-5) = 0 + B(-3)(-2) + 0$$

$$-6 \times 5 = 6B$$

$$\therefore B = -5$$

**Step 5** of 7

$$H(\omega) = \frac{2}{1+j\omega} - \frac{5}{4+j\omega} + \frac{3}{2+j\omega}$$

We know that  $e^{-at}u(t) \leftrightarrow \frac{1}{a+j\omega}$

Using above transform pair apply inverse Fourier transform on both sides of  $H(\omega)$ , we get

$$h(t) = 2e^{-t}u(t) - 5e^{-4t}u(t) + 3e^{-2t}u(t)$$

$$\therefore \boxed{h(t) = (2e^{-t} - 5e^{-4t} + 3e^{-2t})u(t)}$$

**Step 6** of 7

(b) Given Fourier transform

$$Y(\omega) = \pi\delta(\omega) + \frac{1}{j\omega} + \frac{2(1+j\omega)}{(1+j\omega)^2 + 16}$$

We know that the inverse Fourier transform of

$$F^{-1}\left\{\pi\delta(\omega) + \frac{1}{j\omega}\right\} = u(t)$$

$$F^{-1}\left\{\frac{(a+j\omega)}{(a+j\omega)^2 + \omega_0^2}\right\} = e^{-at} \cos(\omega_0 t)u(t)$$

**Step 7** of 7

Using above two equations, apply inverse Fourier transform on both sides of  $Y(\omega)$ , we get

$$F^{-1}\{Y(\omega)\} = F^{-1}\left\{\pi\delta(\omega) + \frac{1}{j\omega} + \frac{2(1+j\omega)}{(1+j\omega)^2 + 16}\right\}$$

$$y(t) = F^{-1}\left\{\pi\delta(\omega) + \frac{1}{j\omega}\right\} + 2F^{-1}\left\{\frac{(1+j\omega)}{(1+j\omega)^2 + 4^2}\right\}$$

$$y(t) = u(t) + 2e^{-t} \cos 4t u(t)$$

$$\therefore \boxed{y(t) = (1 + 2e^{-t} \cos 4t)u(t)}$$